

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1790

Cooks Pond 230 kV
April 30, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1790. The Interconnection Customer proposes a 143.0 MW solar photovoltaic facility interconnecting at the Cooks Pond 230 kV substation in the Ohio area. The request was received on June 22, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 460 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,136,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MECHANICVILLE HYDROELECTRIC STA / CATLETT 100 kV	DRESDEN / GREENLY 500 kV L	126.1	150	7.4
SMITHERS / MECANICVILLE HYDROELECTRIC STA 100 kV L	SMITHERS SUBSTATION / MECHANICVILLE HYDROELECTRIC STATION 100 kV	107.3	220	10.0
MECHANICVILLE HYDROELECTRIC STATION / SMITHERS 100 kV Circuit 2	SMITHERS / MECHANICVILLE HYDROELECTRIC STA 100 kV Line 1	101.2	233	10.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MECHANICVILLE HYDROELECTRIC STATION / CATLETT 100 kV line	Rebuild line with higher-capacity conductor	\$5,063,000	100% (\$5,063,000)
MECHANICVILLE HYDROELECTRIC STA / SMITHERS 100 kV Cir. 1	Reconductor line	\$905,000	66% (\$594,000)
MECHANICVILLE HYDROELECTRIC STATION / SMITHERS 100 kV Circuit 2	Reconductor line	\$730,000	66% (\$479,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$6,136,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.